

Study Questions: Action — Teaching a New Stitch Across the Circle

1. Describe the **teaching loop** in Active Inference terms. What does the teacher predict, observe, and update in each cycle of the loop?
2. What does it mean for a teacher to build a **second-order generative model** — a model of the learner's model? Give a concrete crochet example.
3. When a teacher demonstrates a bobble stitch and the learner fumbles, the teacher adjusts their approach. What **prediction error** has the teacher experienced, and how do they use it?
4. How does the learner's **prediction error** during a stitch attempt serve as the raw material of learning? Why is a supportive circle environment important for making these errors productive?
5. Compare teaching by **demonstration** (showing), by **verbal explanation** (telling), and by **physical guidance** (moving the learner's hands). How does each strategy address prediction error differently?
6. A skilled teacher switches strategies mid-lesson when they notice the learner is still confused. Explain this in terms of **active inference** — what triggered the switch, and what is the teacher trying to achieve?
7. When a circle decides to make a communal blanket, describe the **collective action** involved. What predictions does the group make, and how are individual actions coordinated?
8. What happens when individual contributions to a collaborative project do not match — squares of different sizes, clashing colors? How does the circle handle these **collective prediction errors**?
9. Describe **helping** as an action taken to reduce another agent's prediction error. What does the helper perceive, predict, and do?

10. How does **encouragement** function in Active Inference terms? What is the helper adjusting in the learner's generative model when they say "You are doing great"?
11. When the circle debates whether to attempt a complex lace pattern or a simpler project, they are collectively evaluating **expected free energy**. Explain what this means in practical terms.
12. Think about a **group decision** your circle has made (real or imagined). How did multiple generative models contribute to the outcome? Was there disagreement, and how was it resolved?
13. How does the concept of **action-perception loops** apply to a teaching moment in the circle? Both teacher and learner are simultaneously acting and perceiving — describe the interleaving.
14. A circle member gives a finished hat to a friend outside the group. Analyze this as an **active state** in the circle's Markov blanket. What is being transmitted from inside to outside?
15. How does the circle's culture around mistakes (frogging is normal, errors are expected) create conditions that support active inference in learning? What would happen if errors were stigmatized?
16. Compare the action of a solitary crocheter following a YouTube tutorial with the action of a circle member being taught in person. What advantages does the in-person circle offer for the active inference teaching loop?
17. When a complete beginner joins the circle, the group must act collectively to integrate them. What social actions are involved, and how do they function as active inference?
18. Describe a scenario where the circle's **collective action** fails — a collaborative project goes wrong, a group decision leads to frustration. How does the group use this collective prediction error?
19. How does **distributed execution** in a collaborative project differ from a single person making the whole thing? What are the advantages and challenges of distributing action across multiple agents?

20. Reflect on the relationship between action and all the modules that came before (systems, agents, perception, cognition). How does action depend on each of these? Can there be meaningful action in a circle without the preceding elements?